AN INTRODUCTION

Chain Theory

Toward fundamental physics from distilling networks of entangled qudits

A. D. Wissner-Gross
alexwg@alexwg.org · www.alexwg.org
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THE CLAIMS

What this talk claims, and what it does not

DERIVED

From the axiom and its stated definitions

- One exact number, (79 + 48√2 + 42√3 + 3√6)/31104, whose inverse is 137.035 999 189 566…
- Its computation, checked in a single Lean 4 file

GRADED

A compendium of preliminary findings

- 100 theorems
- 109 conditional theorems, on named conditions
- 198 open results, neither established nor excluded

CONJECTURED

Tested by measurement, not proved

- That the number is α(0), which it matches to 0.6σ of CODATA 2022
- That the six genres of link correspond to sectors of physics
- That the recorded webs are the Standard Model particles

THE PLAN

Six parts

1
Why new foundations

Fifty years without new structure, and three clues

2
The axiom

One sentence, five definitions, one cap, six genres of link

3
One exact number

The alpha Letter: the count, the comparison, the conjecture

4
The Lean checks

The computation, checked in a single Lean 4 file

5
The compendium

What else has been worked out, and how it is graded

6
A sharp test

The next measured digits, and how to check everything

WHY NEW FOUNDATIONS

1900 to 1928: physics rebuilt from its foundations

1900	1905	1913	1915	1925	1928
Planck’s quantum	Special relativity	The quantum theory of the atom	General relativity	Quantum mechanics	Dirac’s equation

Each was a new framework, and most were confirmed within a few years. The decades that followed turned them into the Standard Model, whose structure was considered complete by the mid-1970s.

WHY NEW FOUNDATIONS

Fifty years of confirmation, not new structure

Found, as predicted decades earlier

Discovery	Year
The W and Z bosons	1983
The top quark	1995
The Higgs boson	2012
Gravitational waves	2015

Found, and explained by no theory

Surprise	Year
Neutrino masses	1998
Accelerating expansion	1998

Both added free parameters to a count that was meant to shrink.

The Standard Model settled the structure but not the numbers: its two dozen or so parameters are measured, not derived, and nothing within it explains their values.

WHY NEW FOUNDATIONS

The aim changed, from derivation to selection

Anthropic selection

The constants are environmental, selected by the existence of observers. Its one notable success: Weinberg’s 1987 bound on the cosmological constant, a decade before the accelerating expansion was measured.

The string landscape

A vast population of vacua to select from. Predictions rest on our being typical observers, and what counts as typical depends on a measure no one has fixed.

The swampland

A partial return: which effective theories are consistent with quantum gravity at all. It asks largely from within string theory, and leaves the constants where the landscape put them.

The opposite course: keep derivation as the goal, and look for new foundations.

WHY NEW FOUNDATIONS

Three clues, each read as something else

A bound

read as a fact about black holes

A gauge group

read as a blemish to be embedded

A null result

read as a scale not yet reached

A derivation with no free parameters, and no appeal after the fact to the existence of observers, may need no new experiment to begin.

CLUE 1 · A BOUND

A bounded region holds finitely many degrees of freedom

Result	Source
The entropy bound	Bekenstein 1981
Holography	’t Hooft 1993; Susskind 1995
Gravity from horizon thermodynamics	Jacobson 1995
Spacetime from entanglement	Van Raamsdonk 2010
Quantum theory from information	Hardy 2001; Chiribella, D’Ariano, Perinotti 2011

The lesson for matter

A field has infinitely many degrees of freedom in every region, and a string infinitely many modes. A finite bound says to look instead for finitely many degrees of freedom, carried by quantum systems of finite dimension.

CLUE 2 · A GAUGE GROUP

The gauge group as a signature, not a blemish

$$S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1)) / \mathbb{Z}_6$$

The local basis changes at the two ends of a qutrit maximally entangled with a qubit: SU(3) at one end, SU(2) at the other, and a relative phase that neither end owns.

Embedded, so far without confirmation

SU(5) (Georgi and Glashow, 1974), then SO(10), E_6 and supersymmetric versions. No proton decay has been seen, and no superpartner has appeared.

Read instead as one pair

The group is familiar as the stabilizer of a 3 + 2 split in SU(5) (Baez and Huerta, 2010). The clue is the reading: the symmetry of one entangled pair.

CLUE 3 · A NULL RESULT

No substructure: undetectable, or undistillable?

Every particle behaves as a point

Searches for the excited states and contact interactions that constituents would produce have returned nothing. Yet three generations repeating one set of charges is the kind of pattern that has always signaled constituents, from the periodic table to the hadrons.

Bound entanglement

Certifiably entangled, yet no local operations and classical communication distill a maximally entangled pair from any number of copies (Horodecki, Horodecki and Horodecki, 1998). It first becomes possible when $d_1 d_2 > 6$, and a witness can detect it.

On this reading, the substructure of elementary matter would not be undetectable. It would be undistillable.

WHY NEW FOUNDATIONS

The clues combine into a network

1

Finitely many qudits

Every degree of freedom is discrete: finitely many quantum systems, each of finite dimension.

2

Maximally entangled pairs

A pair entangled any less would carry continuous Schmidt coefficients, which a discrete theory cannot afford.

3

Shared qudits

Pairs join where they share a qudit; by monogamy it holds one copy per pair, matched at the join, as at a PEPS site.

4

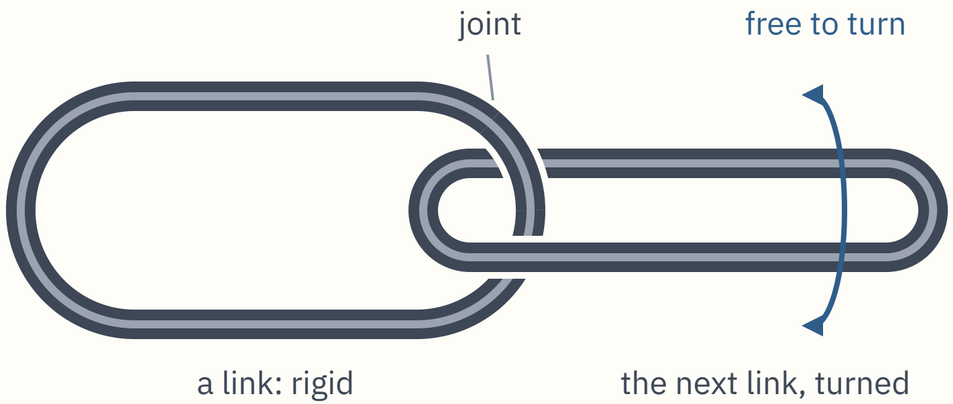
Two kinds of link

Some links are fixed by a single distillation. The rest are just as maximally entangled, but no single distillation fixes them.

A particle is then a finite chainmail of such pairs: a chain, not a string.

WHY NEW FOUNDATIONS

Rigid links, free joints



A chain

Link: rigid, its two ends turn together

Joint: two links meet and turn freely

What no turn of a joint changes

A network of pairs

Pair: a basis at one end fixes the far end’s basis

Shared qudit: bases not matched across its pairs

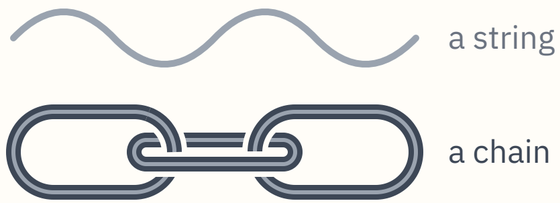
What is known: what no matching changes

IN ONE SENTENCE

Chain Theory studies finite networks of maximally entangled pairs of qudits, joined at shared qudits with no fixed identification of bases across the joins, and known only through what distillation can fix with certainty, which is what every identification leaves unchanged.

WHY NEW FOUNDATIONS

Why “Chain Theory”: the name has three sources



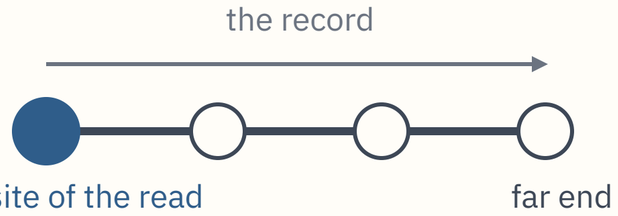
1 Chains, not strings

Elementary particles are postulated to be finite chains or chainmail of discrete links, rather than continuous strings.



2 Free joints

Pairs join at shared qudits, and each join carries a gauge freedom in how bases are matched across it.



3 Read from one end

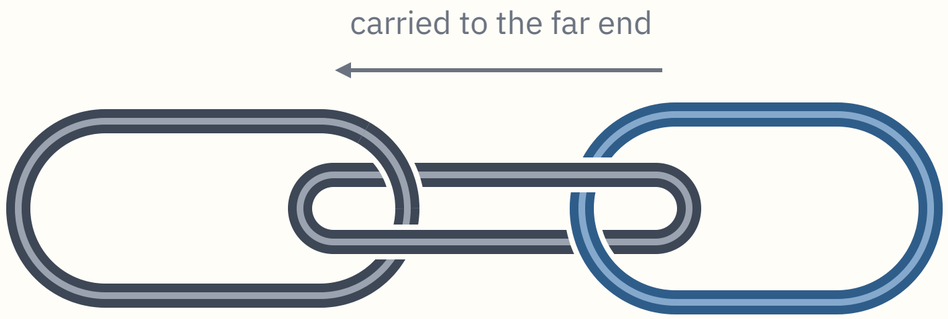
The record has a direction: every walk is read outward, from the site of the read to the far end.

HENCE THE NAME

Finite links, free at the joints, read from one end: a chain, not a string.

WHY NEW FOUNDATIONS

To know is to link





The pairs move from two sides of the square to the other two. Distill 1 with 3 instead, and the diagonals become the pairs.

DISTILLATION

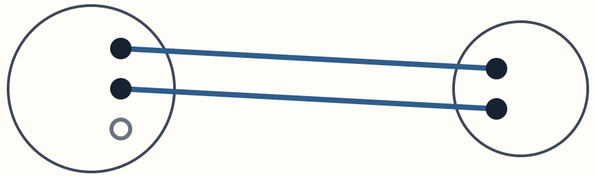
To distill two qudits is to project them onto the maximally entangled state of their common support: a generalized distillation, with one copy, one joint projection, and no failure branch. Between pairs it moves entanglement without adding any, and a single read fails to fix the ends of a pair exactly where standard distillation fails to reach every entangled state, at $d_1 d_2 > 6$.

WHY NEW FOUNDATIONS

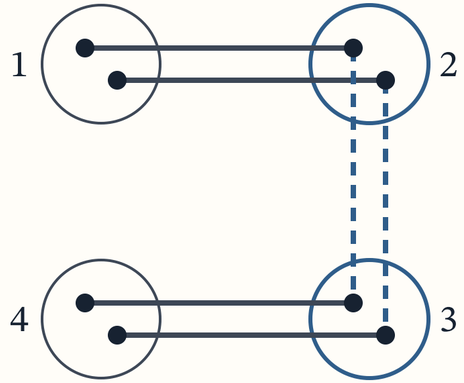
A pair is d strands



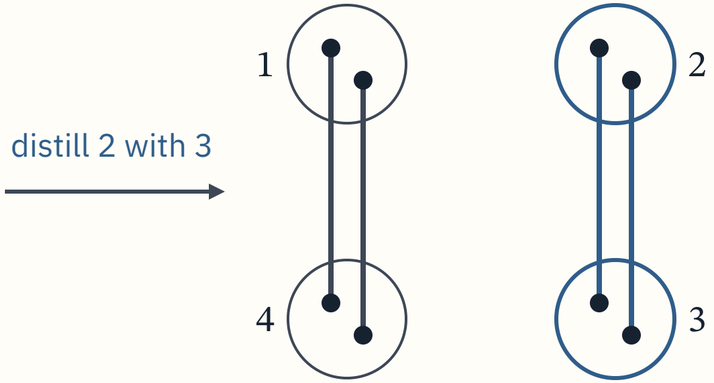
two qutrits: 3 strands, one per term of $|\omega_3\rangle$



a qutrit and a qubit: 2 strands; the third dot is cut off



pairs 1–2 and 3–4; dashed: the projection, 2 to 3



follow each strand: 1 meets 4, dot for dot

Follow the strands. A pair is d strands, one per term of $|\omega\rangle$, each at amplitude $1/\sqrt{d}$. A path from 1 to 4 crosses three strands, picking up $1/(d\sqrt{d})$, and its d paths sum to $(1/d)|\omega\rangle_{14}$.

The same √d as in the count. A surviving record runs over the d strands of its pair at $1/\sqrt{d}$ each, giving $\sqrt{d}$. The dots are only a basis: $U \otimes \bar{U}$ at both ends leaves a pair as it is.

Distillation in bra-ket notation

$$P_m = |\omega_m\rangle\langle\omega_m|, \qquad |\omega_m\rangle = \frac{1}{\sqrt{m}} \sum_{i=1}^{m} |i\rangle|i\rangle, \qquad m = \min(d_1, d_2)$$

The act. Two qudits are projected onto the maximally entangled state of their common support. For a qutrit with a qubit m = 2, and the qutrit's third direction is cut off.

$$P_{23}\,|\omega\rangle_{12}\,|\omega\rangle_{34} = \frac{1}{d}\,|\omega\rangle_{23}\,|\omega\rangle_{14}$$

Linking two pairs. 1 and 4 become a pair, as in entanglement swapping. Ordinary quantum mechanics gives this branch probability $1/d^2$; the axiom admits no other.

$$(A \otimes \mathbb{1})\,|\omega\rangle = (\mathbb{1} \otimes A^T)\,|\omega\rangle, \qquad (U \otimes \bar{U})\,|\omega\rangle = |\omega\rangle$$

What a pair carries. Whatever a read fixes at one end it fixes at the far end; and a pair matches its ends' bases without choosing one, which is the freedom at the joints.

WHY NEW FOUNDATIONS

Acts without a clock

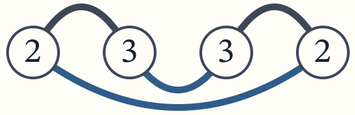
Every act is a distillation. A read is one too, taken at one end of a pair, in a basis of that end.

$$P_{23}\,|\omega\rangle_{12}\,|\omega\rangle_{34} = \frac{1}{d}\,|\omega\rangle_{23}\,|\omega\rangle_{14}$$

with the act — two new pairs

1 One identity

“Before” and “after” are one configuration written two ways, not two moments. Nothing travels to 1 or 4.



$6 \cdot 6$
$4 \cdot 9$

one 36, paired two ways

2 Like factoring 36

Acts on separate qudits commute, so any order gives one network, as any bracketing of 2·3·3·2 gives 36. Unlike factoring, each regrouping takes an act.

$$XZ = -ZX$$

two reads at one qubit

3 Where order is kept

Two reads of one qubit in different bases anticommute, so their order can’t be rewritten away. It is kept, as a sign that returns in the count.

TIME

Time is counted, not assumed: the compendium’s order coordinate counts steps between acts (Box 50.1, conditional).

THE AXIOM

Nothing exists but finitely many maximally entangled pairs of qudits, some qudits shared, known only by distilling them.

The only input. In the spirit of Wheeler’s “it from bit” (1989), with finite quantum systems in place of bits.

THE AXIOM

Five definitions of the axiom's terms

Word	Definition
qudit	a finite-dimensional quantum system, of dimension at least two, since a one-dimensional system ends no pair
shared	a qudit that is an end of two or more pairs, carried as one copy per pair; the copies are one qudit, so exchanging two pairs moves only their far ends
distill	project two qudits of dimensions d_1, d_2 onto $P_m = \|\omega_m\rangle\langle\omega_m\|$ on their common support, $m = \min(d_1, d_2)$; the output is the pair of their type
read	an act at one end of a pair that distills the two ends in a basis of that end, one read for each of its $d + 1$ mutually unbiased bases, and passes the basis along the pair, leaving the far end in $\langle e\|\omega_m\rangle$ at each element $\|e\rangle$, since $(A \otimes 1)\|\omega\rangle = (1 \otimes A^{T})\|\omega\rangle$; the basis reaches the far end whole, so a read is taken at the larger end
known	what its reads fix jointly, under every matching of its pairs and every exchange of its reads or pairs; two anticommuting reads also fix their product, whose sign flips with their order

A pair, or link, of type (d_1, d_2) joins two qudits in the maximally entangled state on their common support, $|\omega_m\rangle = (1/\sqrt{m}) \Sigma_i |i\rangle|i\rangle$. A matching, a change of basis by a unitary U of the support, acts on the pair as $U \otimes \bar{U}$ and leaves it unchanged. To fix is to determine with certainty from a single copy, not to estimate an expectation value over many. Certainty leaves nothing open, which gives the read cap.

The definitions spell out the axiom's terms; their properties follow from it (the Letter's Supplemental Material, Sec. S1).

THE AXIOM

Three consequences settle the measurement theory

The twirl

The pair state is unchanged, or envariant (Zurek, 2003), under every matching, so every readout is invariant under their average, the isotropic twirl.

One copy

The pairs are finitely many and distilling creates no qudit, so a read acts once on the one copy of its link. What matters is what a single read fixes, not asymptotic distillability.

A trace

The operators invariant under every matching are spanned by 1 and P_m (Horodecki and Horodecki, 1999), so a read delivers its pair projector and nothing finer.

$$\langle M \rangle = \mathrm{Tr}_R\, M \,/\, \dim R$$

The readout of a configuration, a set of links some of them read. R holds the whole operator space of each read link and the traceless part of each unread link. The bare insertion, the identity of R, reads one; the dressed insertion M weighs only what the reads fix.

THE DEFINITION OF FIX

Readable or dark: readable exactly when $d_1 d_2 \leq 6$

CERTAINTY LEAVES NOTHING OPEN

A read fixes two ends only when every consistent relation between them, every positive map between their operator algebras, is of a kind distillations carry.

Reads compose only decomposable maps, so a type is readable when every positive map between its ends is decomposable, and dark otherwise. No state is bound entangled: what no distillation carries is a relation, not a state.

Type	Positive maps	Source
(2,2)	all decomposable	Størmer 1963
(3,2)	all decomposable	Woronowicz 1976
(4,2)	some not decomposable	Woronowicz 1976
(3,3)	some not decomposable	Choi 1975

Non-decomposable maps embed in every $d_1 d_2 \geq 8$, and no product of integers ≥ 2 is 7.

The read cap. The readable types are (2,2) and (3,2), of which (3,2) alone has unequal ends. The others are dark.

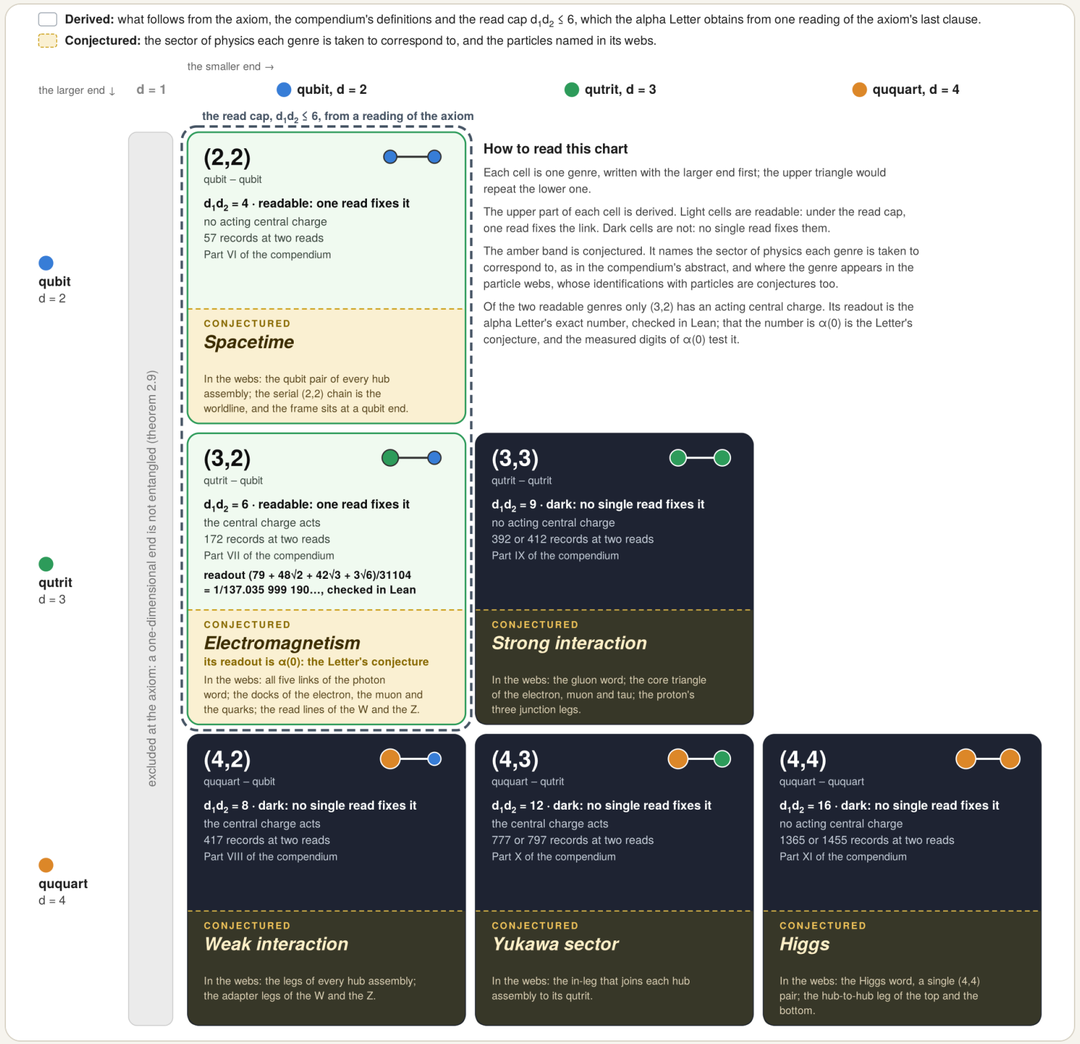
THE SIX GENRES

Six genres of link, two of them readable

A link’s genre is the pair of its end dimensions, written larger end first. Under the read cap, (2,2) and (3,2) are readable; the other four are dark.

The upper part of each cell is derived. The amber band is conjectured: the sector of physics each genre is taken to correspond to.

That ends have dimension two, three or four is the compendium’s box 1.7, “Conjecture 0”, graded a conditional theorem under its move table. A one-dimensional end is excluded at the axiom (theorem 2.9).



ONE EXACT NUMBER

The sharpest test is a number

Measured, not derived

α(0) is known to about one part in ten billion, one of the most precisely measured numbers in science, yet the Standard Model takes it from experiment.

Hard to hit by accident

Its digits show no pattern. A simple value like the 2/3 of Koide's lepton-mass relation can be reached by many short formulas. Ten digits of 137.035 999… cannot.

Nothing left to tune

Proposed closed forms foundered on their normalization: a volume can be rescaled, and the number with it (Wyler 1969, 1971; Robertson 1971; Gilmore 1972). A derivation must leave nothing to fix.

"all good theoretical physicists put this number up on their wall and worry about it"

"a magic number that comes to us with no understanding by man"

Richard Feynman, *QED: The Strange Theory of Light and Matter* (1985)

ONE EXACT NUMBER

What must be computed: the on-shell coupling

The on-shell scheme

For a massless spin-one field coupled to a conserved abelian charge, the coupling at zero transfer is universal (Weinberg 1964, 1965). The Ward identity $Z_1 = Z_2$ reduces it to the vacuum polarization, and charged states of any mass decouple from this limit (Appelquist and Carazzone 1975).

$$\alpha(0) = Z_3 \, \alpha_bare$$

So the model must supply

1. one object with a conserved abelian charge obeying a Ward identity
2. a two-point function with its pole at zero transfer
3. its dressed-over-bare two-point function at that pole

ONE EXACT NUMBER

The central charge picks one readable genre

The identification algebra of a link, $su(d_1) \oplus su(d_2) \oplus u(1)$, is the infinitesimal basis changes at its two ends. Its center is spanned by

$$Q = i\,\mathrm{diag}(d_2\ 1, -d_1\ 1)$$

which acts on the whole link, its pair ket included, as the scalar $i(d_2 - d_1)$: zero at (2,2), and i at (3,2).

At (3,2)

- The algebra is $su(3) \oplus su(2) \oplus u(1)$, of $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$
- $Q = 6iY$, with $Y = \mathrm{diag}(\tfrac{1}{2}, \tfrac{1}{2}; -\tfrac{1}{3}, -\tfrac{1}{3}, -\tfrac{1}{3})$ the hypercharge generator
- The link carries (3,2) with Y = 1/6, the quantum numbers of the left-handed quark doublet
- The pair's stabilizer has dimension four, its center $U = i\,\mathrm{diag}(1, 1; -1, -1, 0)$

So exactly one readable genre carries an acting central charge. Its charge is the hypercharge direction; the passage to electric charge lies beyond the Letter.

ONE EXACT NUMBER

Three conditions leave one configuration

THEOREM · UNIQUENESS

Among the configurations of readable links at a shared qudit, exactly one is a two-point function, is read at the qutrit end, and carries an acting central charge on every link: a shared qutrit with three links to qubits, two of them read.

Acting charge

(2,2) links carry none, since Q vanishes on their pair, so only (2,3) links remain.

End rule

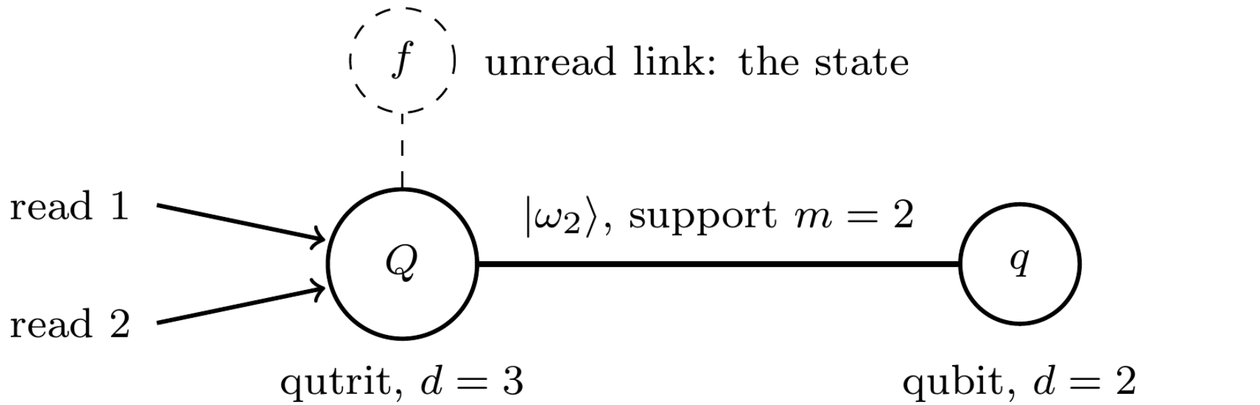
A read reaches its far end whole, so (2,3) links are read at the qutrit end, and the shared qudit is a qutrit.

Two-point function

Two read links and one unread link of the same type: three qubit links, two of them read.

ONE EXACT NUMBER

The object: a qutrit–qubit link, read twice

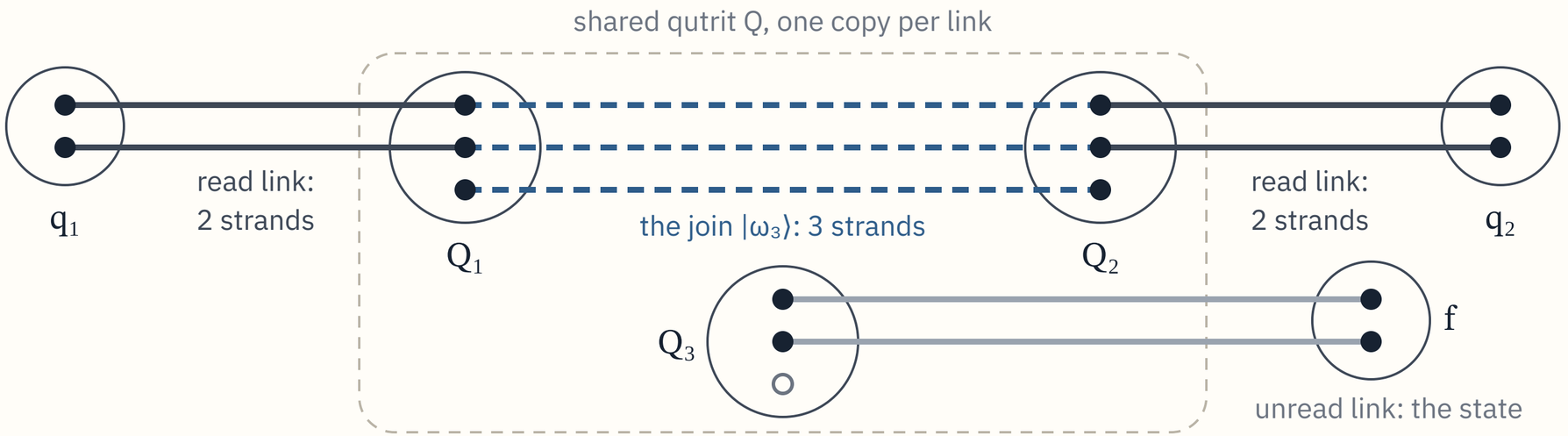


Two reads at the shared qutrit Q, once on each of two links of this type, drawn as one.

A third link of the same type, to the qubit f, is left unread: reading it would be a third insertion. It carries the state.

This is the model's definition of a two-point function.

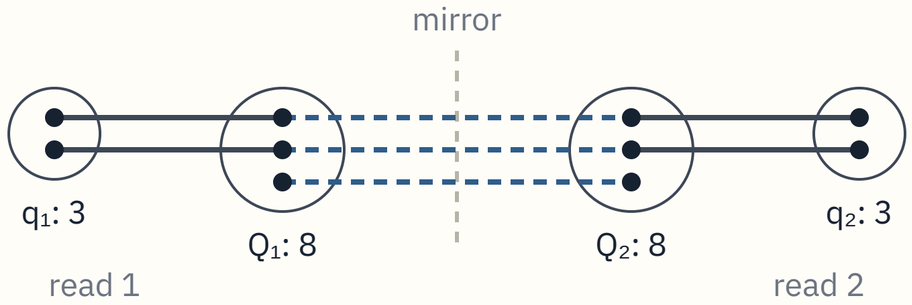
The object, as strands



Follow the strands. From either dot of q_1, a strand runs through Q_1, across the join and through Q_2 to q_2: two paths of $1/(2\sqrt{3})$, together $1/\sqrt{6}$, the Supplement's $\sqrt{6}$.

Four slots. Letters will sit at the four ends of the two read links: the copies Q_1 and Q_2, and the read qubits q_1 and q_2. The unread link carries the state.

The counts: fold the picture along its mirror

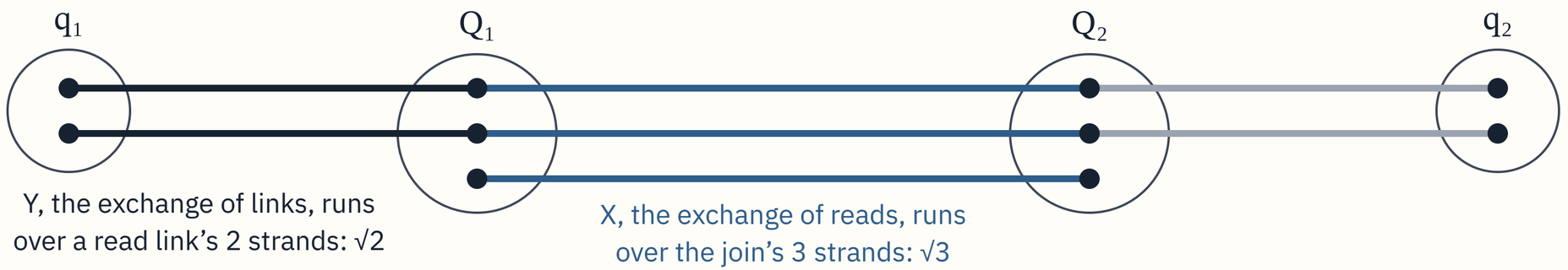


The fold. A placement and its mirror image coincide or pair up. A pair gives one symmetric and one antisymmetric record, a self-image one symmetric record: n placements with f self-images give (n + f)/2 symmetric and (n – f)/2 antisymmetric.

Boxes. A letter is a box on the strands at a slot: $3^2 - 1 = 8$ at a copy, $2^2 - 1 = 3$ at a qubit. A placement is two boxes at two different slots: $8 \cdot 8 + 3 \cdot 3 + 4 \cdot (8 \cdot 3) = 169$. The mirror swaps read 1 and read 2, and 11 placements are their own image: the same box on both copies (8) or on both qubits (3).

Placements	Count	Survives
All 169, antisymmetric part	(169 – 11)/2 = 79	neither
Same kind of end (73), symmetric part	(73 + 11)/2 = 42	X
Copy with qubit (96), symmetric part	96/2 = 48	Y
Commutators at the qubits	3	both

The weights: the strands each exchange runs over



Why √d. Each strand carries 1/√d. A record that survives an exchange runs over all d strands where that exchange acts: d · 1/√d = √d.

Both, and neither. The 3 commutators survive both exchanges: √3 · √2 = √6. The 79 antisymmetric records survive neither and weigh 1.

The qubit–qubit records. Their boxes sit at the qubits, yet X still runs over the join, so they take √3, since by the Letter's definition a read is an act at the end where it is taken.

IN STRANDS · 4 OF 5

All 172 records, each its own strand diagram



weight 1 · 79 √2 · 48 √3 · 42 √6 · 3

Straight jumper: symmetric letter · curved: antisymmetric · filled and hollow dots: diagonal · + or –: with the mirror image

The denominator, and the whole number

24

The unread link. A letter at each end, no blanks: 8 × 3. This is its connected part, the state.

× 36

Read link 1. Any operator at each end, blank included: 9 × 4.

× 36

Read link 2. The same: 9 × 4.

Bare and dressed. Every way to decorate the object is one of N = 24 · 36 · 36 = 31104 directions. The bare insertion, the identity, weighs each of them once. The dressed insertion weighs only the 172 records, each by the strands it runs over.

$$\frac{G_{\text{dressed}}}{G_{\text{bare}}} = \frac{79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}}{24 \cdot 36 \cdot 36} = \frac{1}{137.035\,999\,189\,566\ldots}$$

ONE EXACT NUMBER

The readout dimension follows from the two ends alone

$$N = 24 \cdot 36 \cdot 36 = 31104$$

$24 = (3^2 - 1)(2^2 - 1)$

The traceless parts of the unread link: the connected part of its pair projector, which is the state.

$36 = (3 \cdot 2)^2$

The whole operator space of the link at each of the two reads.

The end rule

A read reaches its far end whole. A qutrit-end read fixes the qubit at every basis element; a qubit-end read would miss the qutrit’s third direction. So reads are taken at the qutrit.

ONE EXACT NUMBER

172 measurement records

169

products of two letters

Each read places a letter, a Gell-Mann matrix at the qutrit (8) or a Pauli matrix at the qubit (3), at either end of its link. Each end is a slot, qutrit then qubit for each read: slots (8, 3, 8, 3). A record carries two letters at two different slots, since two at the same slot multiply to a single insertion: 24 + 24 + 64 + 24 + 24 + 9 = 169, of which 48 sit at the two ends of one read.

+ 3

commutators

At a qubit two reads in different bases anticommute. By the definition of the known their commutator is a further record, its sign the order of the reads.

= 172

records

They span the joint record space of the two reads.

Exchanging the two reads fixes 11 products and pairs the other 158 into 79 odd and 79 even combinations. The exchange sorts the record space into its two eigenspaces and keeps both.

ONE EXACT NUMBER

Two commuting exchanges set the weights

X, the exchange of reads

Acts on records with two letters of one type: 73 products and the 3 commutators. A read is an act at the end where it is taken, so X acts at the qutrit wherever the letters sit, and it carries 3.

Y, the exchange of links

Leaves the shared qutrit in place, its copies being one qudit, and swaps the two read qubits: 96 products with one slot at each end, and the 3 commutators. It carries 2.

Survival, and √d

By the definition of the known, a record survives when the exchange returns it times its own sign. Its letter then runs over the d matched directions at amplitude 1/√d: d · 1/√d = √d.

$$k_r = \tfrac{1}{2}\,(2 - X_r)(3 - Y_r)$$

A factor 3 when X reads −1, and 2 when Y reads −1. A record enters with weight $\sqrt{k_r}$, not k_r: it carries the two insertions as two letters, while k_r, four letters, is a four-point quantity.

ONE EXACT NUMBER

The 172 records by the eigenvalues of the two exchanges

Records	(X, Y)	Number	k_r	Weight
odd	(+, +)	79	1	1
even, one slot at each end	(+, −)	48	2	√2
even, both slots at one end	(−, +)	42	3	√3
commutators	(−, −)	3	6	√6
all		172		

The sum of the weights is the numerator of the readout: 79 · 1 + 48 · √2 + 42 · √3 + 3 · √6.

THEOREM · VALUE

The dressed-over-bare two-point function of the object, per unit of its conserved count, is

$$(79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}) / 31104$$

ITS INVERSE

137.035 999 189 566 155 029 183 744…

AN ELEMENT OF ℚ(√2, √3)

with rational coordinates (79/31104, 1/648, 7/5184, 1/10368)

CHECK IT BY HAND

Two ways to redo the count with pencil and paper

WAY 1 · FOUR STEPS

Term by term

Draw every record as an edge of a square whose corners are the four slots. Pair the records with a mirror, weigh each one, and add.

WAY 2 · TWO STEPS

By symmetry

Get the same four numbers, 79, 48, 42 and 3, from the sizes of symmetric and antisymmetric squares of 8 and 3, without listing a single record.

AND THE MACHINE

By computer

The Lean file and a Python script build all 172 records from explicit matrices, the commutators included.

All three use the same rules as Table I, so their agreement checks the arithmetic, not the rules. The rules come from clauses of the Letter’s definitions (Table S1 of its Supplemental Material); the alternatives slide shows what each gives when worded otherwise.

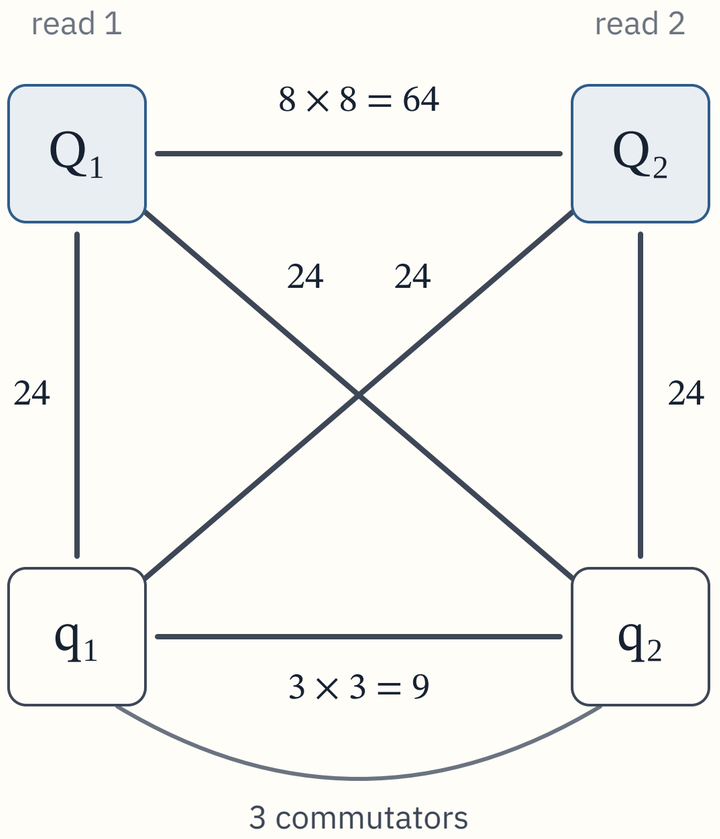
Four slots, and records as edges

Draw the object's two reads (Fig. 1) side by side. The four **slots** are their ends: Q_1 and q_1 for read 1, Q_2 and q_2 for read 2, with Q_1 and Q_2 the two copies of the one shared qutrit.

Letter: an element of a real traceless basis at a slot, $d^2 - 1$ of them at an end of dimension d: 8 at a qutrit, 3 at a qubit (the Gell-Mann and Pauli matrices, up to factors of i).

Record: two letters at two different slots, the two insertions of the two-point function, drawn as an edge. Edges are pairs of slots, not links.

An edge holds the product of its corners' letters: $64 + 9 + 4 \times 24 = 169$. At a qubit two reads in different bases anticommute, so by the definition of the known each of the 3 pairs of qubit letters adds a commutator record, with the order of the reads as its sign: 172 in all.

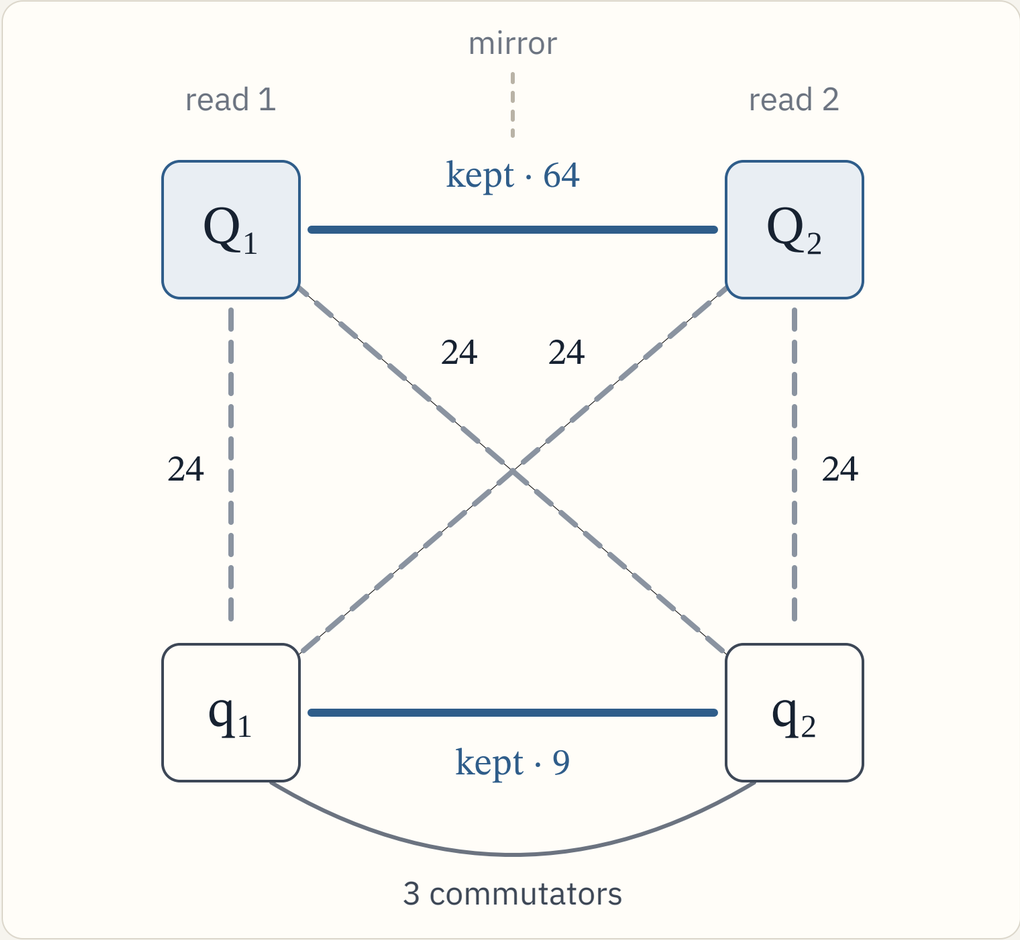


A mirror pairs the records

Mirror: swap the two reads, $Q_1 \leftrightarrow Q_2$ and $q_1 \leftrightarrow q_2$. It keeps the solid edges and swaps the dashed ones in pairs.

Symmetric, antisymmetric: a record r and its mirror image r′ give r + r′ and r − r′. A record that is its own image, the same letter at Q_1 and Q_2 or at q_1 and q_2, is symmetric by itself.

Edge	Records	r + r′	r − r′
Top, kept	64	8 + 28 = 36	28
Bottom, kept	9	3 + 3 = 6	3
Sides	24 + 24	24	24
Diagonals	24 + 24	24	24
Total	169	90	79

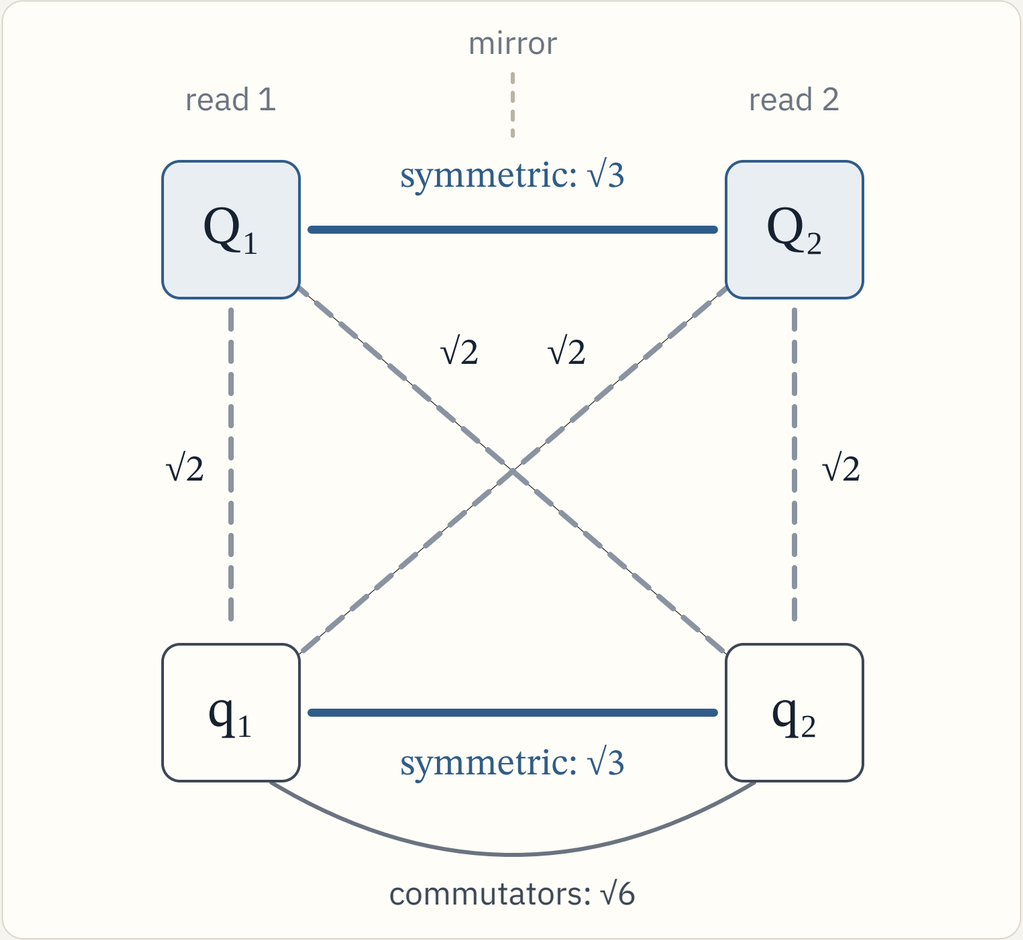


Weigh each record

Every antisymmetric record weighs 1. A symmetric record weighs √3 on a solid edge and √2 on a dashed edge, as marked; each commutator weighs √6.

Why: a symmetric record **survives** the mirror, and its letter then runs over the d matched directions of a pair at amplitude 1/√d each, giving d · 1/√d = √d. On solid edges the mirror is the Letter's exchange of reads, at the qutrit (d = 3); on dashed edges, its exchange of links, which moves the qubits (d = 2). Commutators survive both.

Note the bottom edge: its letters sit at the qubits, yet it takes √3, because a read is an act at the end where it is taken, the qutrit. Worded the other way, α^{-1} would be 139.03.



WAY 1 · TERM BY TERM · STEP 4 OF 4

Add, and divide by the readout dimension

Edge	× 1	× √2	× √3	× √6
Top	28		36	
Bottom	3		6	
Sides	24	24		
Diagonals	24	24		
Commutators				3
Total	79	48	42	3

THE SUM

$$79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}$$

$$= 79 + 67.88225 + 72.74613 + 7.34847$$

$$= 226.97685$$

Divide by the readout dimension, N = 24 · 36 · 36 = 31104:

$$226.97685 / 31104$$

$$= 1 / 137.0360$$

Square roots to six places give 137.0360; to nine, 137.035999.

Symmetric and antisymmetric squares

Take n directions e_1, …, e_n, a basis of a space V. Ordered pairs e_ie_j, one direction in each place, number n^2 and span V ⊗ V. Swapping the two places sorts them into two kinds.

Sym²n: the symmetric combinations, e_ie_i and $e_ie_j + e_je_i$ for $i < j$. There are $n(n + 1)/2$.

Λ²n: the antisymmetric combinations, $e_ie_j - e_je_i$ for $i < j$. There are $n(n - 1)/2$.

Here Sym²n and Λ²n name these counts: Sym²8 = 36, Λ²8 = 28, Sym²3 = 6, Λ²3 = 3. Two different sets of m and n directions, swapped into each other, give m · n of each kind.

Example: three directions x, y, z

xx	xy	xz
yx	yy	yz
zx	zy	zz

Swapping the places reflects the grid across its diagonal: the 3 diagonal cells stay put, and the other 6 meet in 3 pairs.

Symmetric, 6: xx, yy, zz, xy + yx, xz + zx, yz + zy

Antisymmetric, 3: xy − yx, xz − zx, yz − zy, the three components of a cross product

The four numbers from 8 and 3

Weight	Records	Count
1	antisymmetric, on every edge	$\Lambda^2 8 + \Lambda^2 3 + 8 \cdot 3 + 8 \cdot 3 = 28 + 3 + 24 + 24 = 79$
$\sqrt{2}$	symmetric, on the swapped edges	$8 \cdot 3 + 8 \cdot 3 = 48$
$\sqrt{3}$	symmetric, on the kept edges	$\mathrm{Sym}^2 8 + \mathrm{Sym}^2 3 = 36 + 6 = 42$
$\sqrt{6}$	commutators of two qubit letters	$\Lambda^2 3 = 3$

With $a = 3^2 - 1 = 8$ and $b = 2^2 - 1 = 3$ the four counts are $\frac{1}{2}(a^2 - a + b^2 - b) + 2ab$, $2ab$, $\frac{1}{2}(a^2 + a + b^2 + b)$ and $\frac{1}{2}b(b - 1)$: the Supplement's lemma on the sector sizes. The numerator is again $79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6}$.

The rules are those of way 1, applied to dimensions instead of lists, so no record can be dropped or counted twice. What it checks is the counting, not the rules.

Only the dimensions enter. That under SU(3) the 36 is $1 + 8 + 27$ and the 28 is $8 + 10 + \overline{10}$ plays no role, and the formula is for this object, not for other links.

ONE EXACT NUMBER

Why this is the on-shell coupling

$$G = Z_3\,(Z_2/Z_1)^2\,\alpha_bare$$

On-shell datum	In the model
α_bare	1, the bare unit
the residue	normalized by the unit of the readout: *Q* reads one where 1_R reads one
Z_2/Z_1	1, by conservation of the count
Z_3	Tr_R M / N
decoupling	N depends only on the link’s own two dimensions

One definitional identification

Identifying the normalized readout with the on-shell two-point function names what Theorem 2 computes. It is used in no proof.

The model supplies the residue the scheme normalizes and no propagator; the pole’s place at zero transfer enters with the conjecture.

ONE EXACT NUMBER

What else the count could have given

Alternative	1/α	Excluded by
the Letter’s definitions	**137.04**	—
the two ends swapped	135.89	the end rule
both exchanges at the qubit end	146.52	the end rule
both exchanges at the qutrit end	127.54	the qutrit’s copies are one qudit
exchange of reads where the letters sit	139.03	a read acts at its own end
qubit–qubit records moved by Y	134.49	they occupy the same qubits after it
commutators surviving X alone	138.35	Y moves them between the read qubits
weights k_r in place of $\sqrt{k_r}$	97.51	four letters, a four-point quantity
read ends in place of units of charge	70.19	the readout is per unit of the count
no weights	180.84	the factor √d at a passed end

Each is excluded by a definition, without comparison with α. Two land within one percent.

THE CONJECTURE

The identification with α(0) is a conjecture

CONJECTURE · PHOTON

The qutrit–qubit link of Theorem 1 is the elementary link of the photon, and its dressed-over-bare readout is the electromagnetic coupling at zero momentum transfer, α(0).

Why it is not a theorem

The object’s charge is the hypercharge direction. Its passage to electric charge, together with spacetime and helicity, lies beyond the Letter.

What is identified

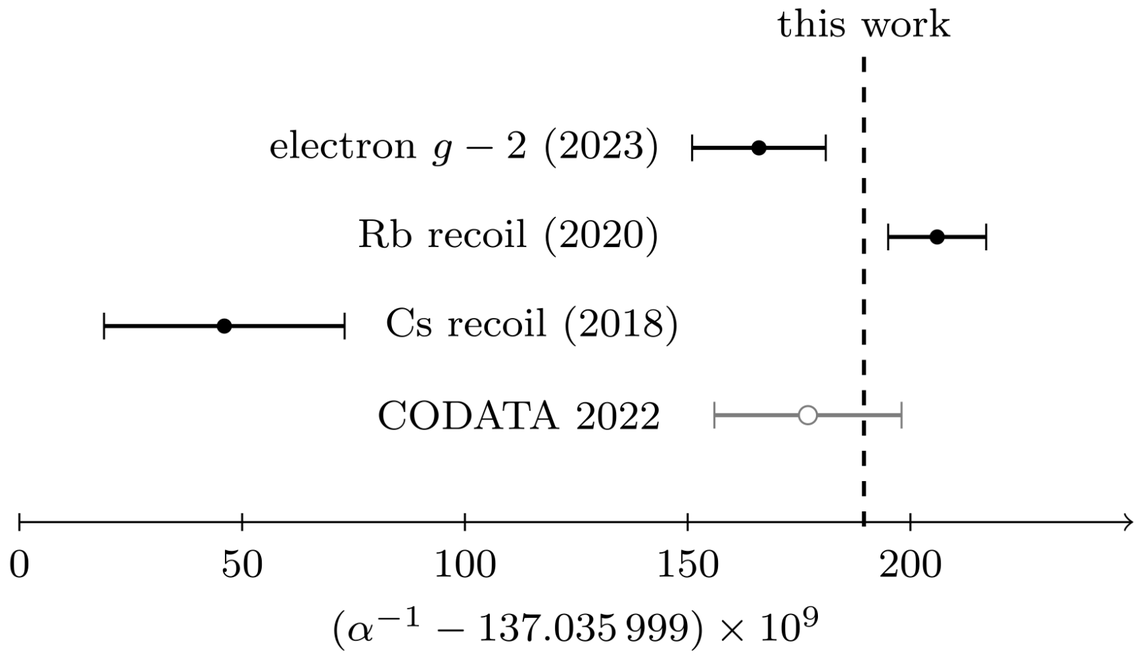
α itself, the unit-free coupling per unit charge, and not e^2 in a convention that carries 4π.

What tests it

Each further measured digit of α(0), including those that settle the tension between rubidium and cesium recoil.

THE CONJECTURE, TESTED

0.6σ from CODATA 2022, between the two most precise



Determination	Distance
electron g – 2, 2023	+1.6σ
Rb recoil, 2020	−1.5σ
Cs recoil, 2018	+5.3σ
CODATA 2022	+0.6σ

The exact form carries no uncertainty of its own. The cesium and rubidium values disagree with each other, an unresolved tension.

The Letter’s Figure 3 · Fan et al. 2023; Morel et al. 2020; Parker et al. 2018; Mohr et al. 2025 (CODATA 2022)

ONE EXACT NUMBER

Every input beyond the axiom

Input	Status
five definitions	the axiom’s terms; properties derived from it
a read fixes two ends only when every positive map between them is decomposable	the definition of fix; it gives the read cap
Størmer, Woronowicz, Choi; Horodecki	theorems of the literature
a read is taken at its own end and reaches the far end whole; a shared qudit’s copies are one qudit; the known holds under every exchange	clauses of the definitions
a record is two letters, with weight $\sqrt{k_r}$	a step of the model
no failure branch of a distillation (in a laboratory, a postselection)	a step of the model, from the axiom
the two-point function (two reads, one unread link) and its units	definitions of the computed quantity
the on-shell correspondence of the readout	one definition; it names the result
the Photon Conjecture	conjecture: the identification with $\alpha(0)$

The read cap, from the definition of fix, selects the link, the on-shell definition names the result, and the conjecture identifies it. Nothing is posed.

THE LEAN CHECKS

The computation, checked in a single Lean 4 file

What the file does

- Uses the core library only: no Mathlib, no imports
- Builds the 172 records from the slots and the commutator matrices; nothing is entered as a number
- Shows each record is an eigenvector of the exchange of reads, and applies the weight law
- Evaluates the readout exactly

```
readout objectWorld objectTwoPoint
= ([(1, 79), (2, 48), (3, 42), (6, 3)], 31104)
```

What it does not check

- Whether the definitions are the right physics: it checks that they produce the number
- The read cap enters only the theorems that select the link, as a named hypothesis citing Størmer, Woronowicz and Choi
- The Photon Conjecture appears nowhere in it

The main theorem has no hypotheses beyond the file's definitions and depends on propositional extensionality alone. A full check takes about two minutes and 2 GB.

THE COMPENDIUM

A compendium of preliminary findings, graded

Release 0.1, a working draft: sixteen Parts, appendices and 52 instrument chapters. Every boxed statement carries its grade, and no measurement enters as an input.

100

theorems, 64 of them tied to named Lean declarations

109

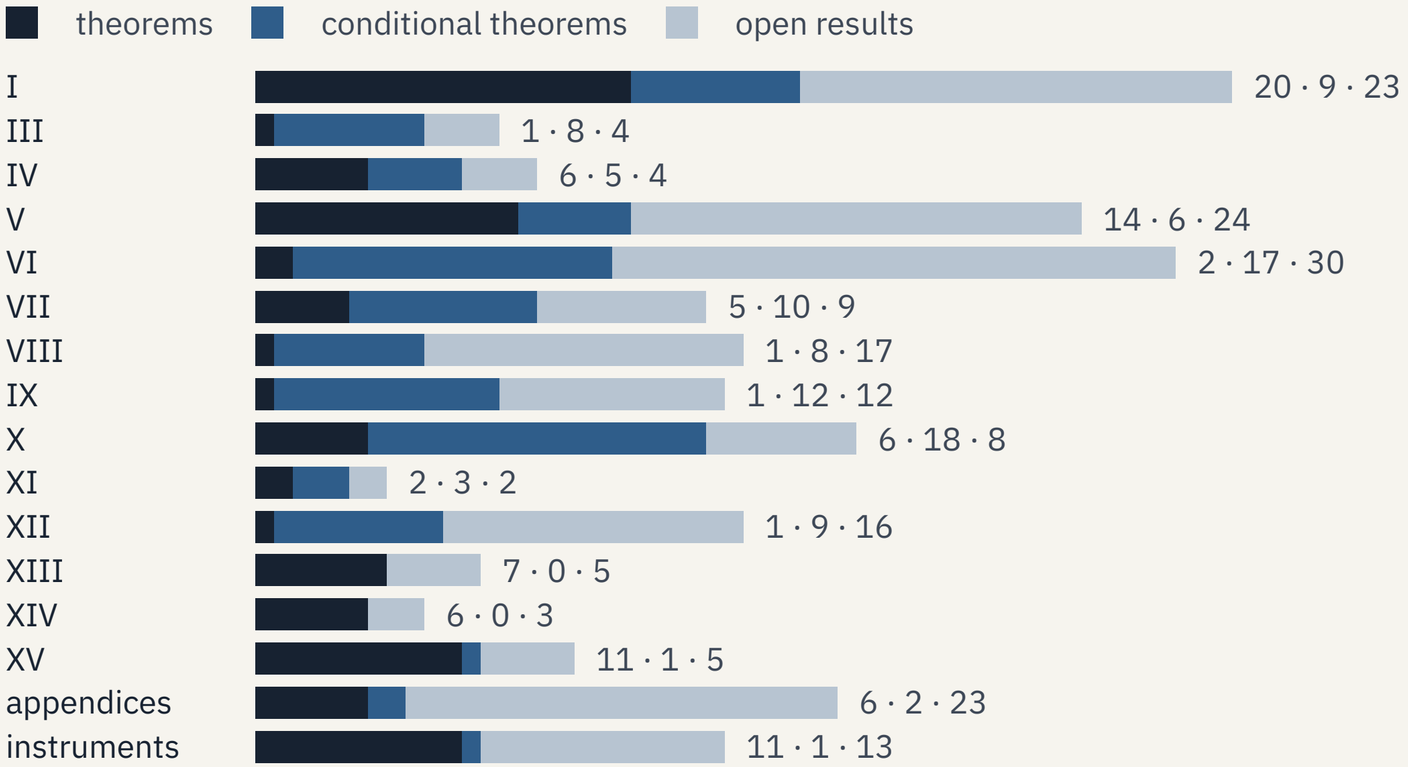
conditional theorems, each with its conditions named in its box

198

open results, each carried to a named step

THE COMPENDIUM

The grades, Part by Part



The six cells

Parts VI to XI, one per genre, hold 17 theorems, 68 conditional theorems and 78 open results.

The most theorems are in the framework (Part I) and the gauge structure (Part V), then quantum information (Part XV).

Parts II and XVI hold tables only.

Counts generated from the compendium’s boxed statements by its index tool · totals 100 · 109 · 198

THE COMPENDIUM

How to read a grade

Theorem	Derived from the axiom and its definitions, by the derivation given in its chapter.
Conditional theorem	Derived under conditions named in its box. A composed claim carries the lowest grade in its chain.
Open result	Stated, and carried to a named step that remains open: neither established nor excluded.

A caution about the word theorem

Many theorems are exhaustive checks over the webs the compendium records. The second law at the counting face, for example, is checked at all 228 one-act reads of fourteen bodies: a census of a finite catalog, not a proof for every web.

THE COMPENDIUM

Organized by the genres of the link

Parts	Genre	Subject, with its conjectured sector	Theorems · cond. · open
I–V	—	foundations: from the framework to the gauge structure	41 · 28 · 55
VI	(2,2)	metric, worldline, horizon · spacetime, conjectured	2 · 17 · 30
VII	(3,2)	the photon · electromagnetism, conjectured	5 · 10 · 9
VIII	(4,2)	carriers and breaking · the weak interaction, by name	1 · 8 · 17
IX	(3,3)	the junction · the strong interaction, by name	1 · 12 · 12
X	(4,3)	the Yukawa leg · the Yukawa sector, by name	6 · 18 · 8
XI	(4,4)	the Higgs and its hub register · the Higgs, by name	2 · 3 · 2
XII–XV	—	applications: from decays to quantum information	25 · 10 · 29
XVI	—	the scorecard: tables only	—

Quantum mechanics, read off the framework

Result	Grade	Box
The state space of a pair type is the span of the act outputs	conditional	35.1
The flat pair and the unitary transport, join and step	conditional	40.1
The uncertainty relation from the Weyl relation	conditional	41.1
Indistinguishability from the twirl	conditional	43.1
Energy as the first-power operator and the step	conditional	44.1
The Wigner function and the many-body sign	conditional	46.1
Measurement as the wash at one act	**theorem**	28.1
The classical limit as the zero-density register	**theorem**	28.5

In 35.1 the inner product of the span is the Born readout: a qudit is its frame, and superposition is the span of the frames. Here the Born rule is an output, conditional on its premises, not an input.

Spacetime and symmetry, from finitely many pairs

Result	Grade	Box
The order coordinate as a count of one-steps	conditional	50.1
The rotation group as the automorphisms of the derived directions	conditional	53.1
The Lorentz group as the isometries of the Hermitian split	conditional	54.1
One against three at every step of the chain	**theorem**	55.1
The spin-c group at the qubit pair	**theorem**	58.1
Parity as the improper element	conditional	59.1
Three involutions and the identity product	**theorem**	61.1

At the qutrit–qubit link, the isometry algebra of the Hermitian split, a form of signature one against three, is six-dimensional and equals the Lorentz algebra at every structure constant. Nothing in the axiom mentions space.

Names from the literature are correspondences by name; each box is a statement about the networks, at the grade shown

The gauge structure, and the number three

Result	Grade	Box
Gauge invariance as the known clause	**theorem**	64.1
The gauge group as the face group of the tether	**theorem**	64.2
The gauge group as a corollary of one construction	**theorem**	65.1
The conserved functionals of the full move set	**theorem**	74.1
The Ward identity from conservation as one link	conditional	78.1
The anomaly with every body at its own sink	**theorem**	79.1
Three, a theorem of the valence	**theorem**	83.1

64.2: the face group of the qutrit–qubit tether is $S(U(2) \times U(3))$, isomorphic to $(SU(2) \times SU(3) \times U(1))/\mathbb{Z}_6$ by an explicit covering. 83.1: the generation count is three at the three core sites of the closed triangle.

These are statements about the networks. That this group is nature’s gauge group, and these three are nature’s generations, belongs to the conjectured correspondence.

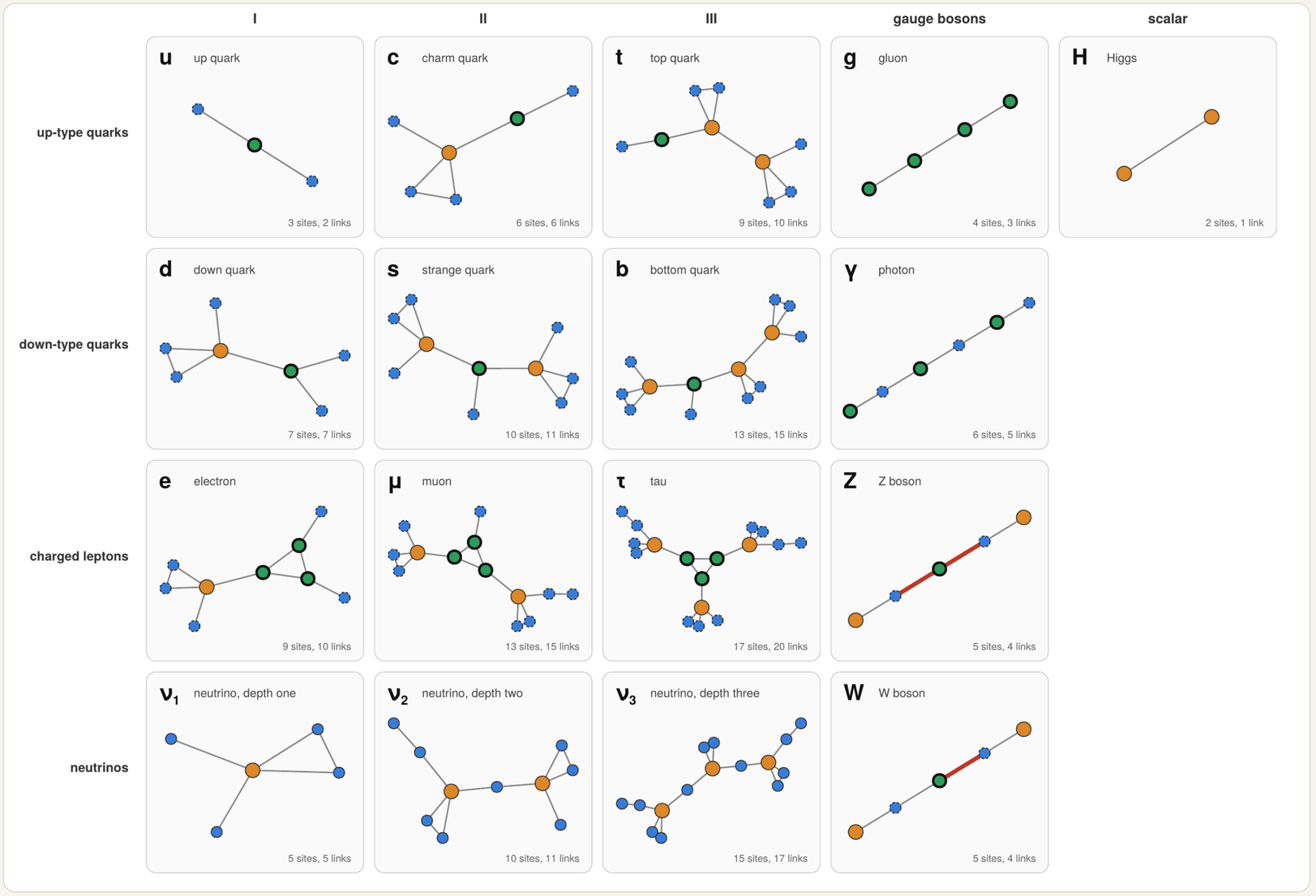
Names from the literature are correspondences by name; each box is a statement about the networks, at the grade shown

CONJECTURED WEBS

Webs for the Standard Model particles

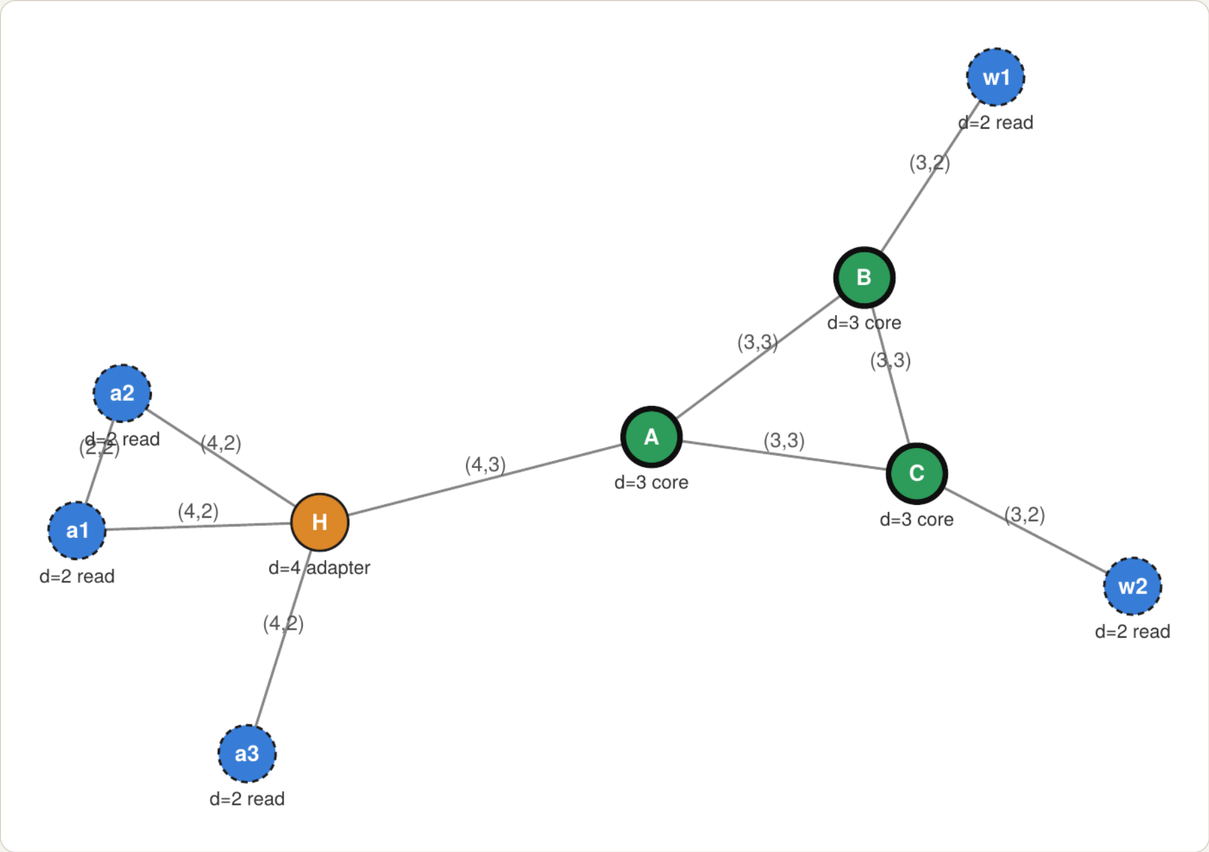
Each site is a qudit and each line a maximally entangled pair. Each generation adds one hub assembly; the W and the Z are one web.

That a web is the named particle is a conjecture. Of the 17 particles drawn, on 16 webs, 1 rests on a theorem, 11 on conditional theorems, 2 on open results and 3 on results not in this release.



CONJECTURED WEBS

The electron’s web, link by link

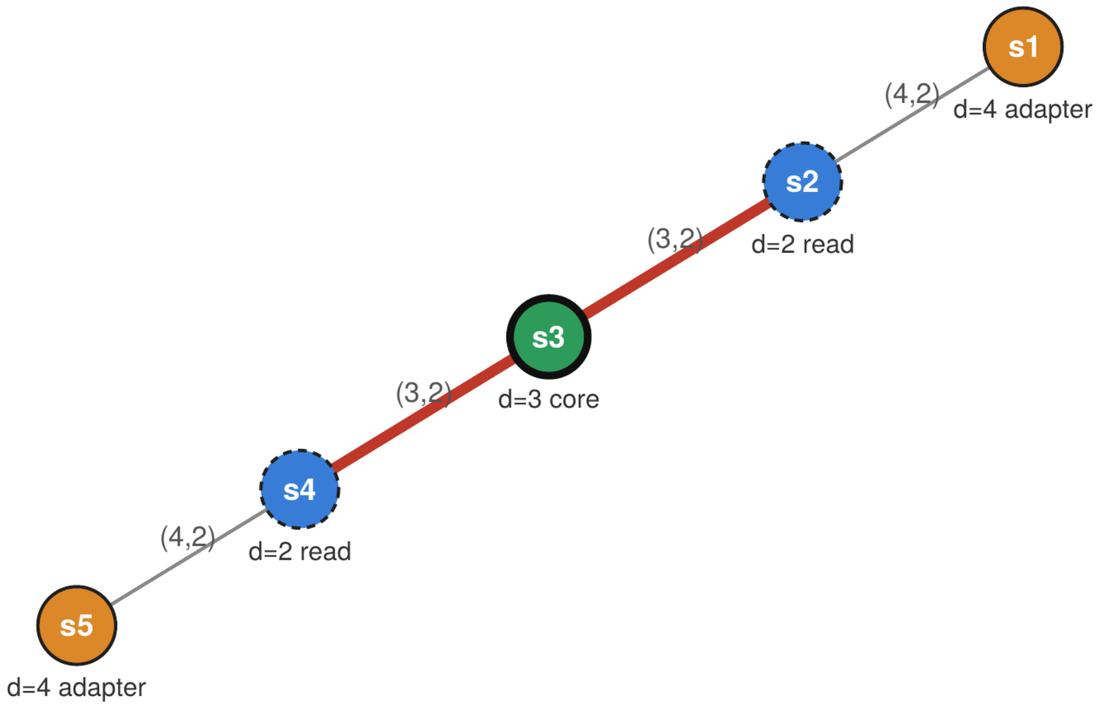


Genre	In the electron
(3,3)	the core triangle of three qutrits
(4,3)	the in-leg joining the hub to the core
(4,2)	the legs of the hub assembly
(2,2)	the qubit pair of the hub assembly
(3,2)	the docks at the other two corners

The muon carries two hub assemblies and the tau three: the lepton ladder, a conditional theorem (152.1), with the tau’s third hub a theorem (154.1).

CONJECTURED WEBS

The W and the Z: one web, two reads



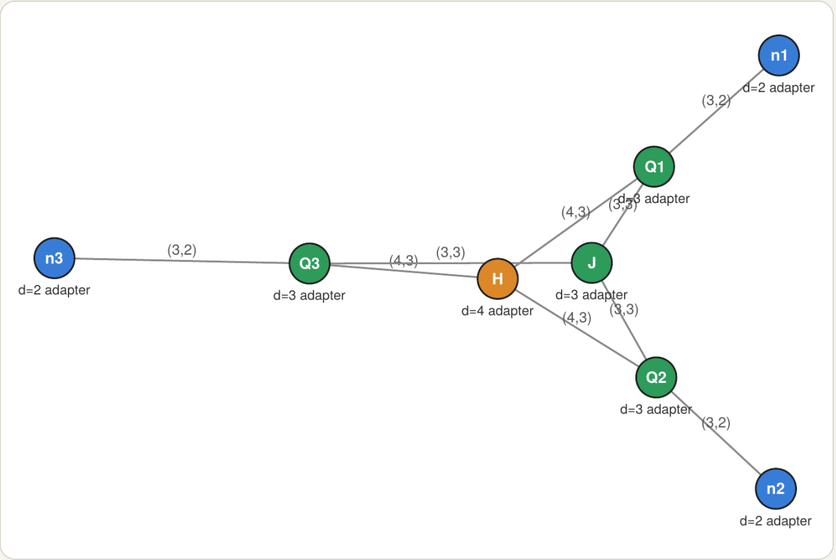
What distinguishes them is the read: the W is read along one line, the Z along two, drawn heavy.

Result	Grade
the W word as the passage web of the cross read	open, 159.2
the Z as the two-line read	open, 159.1
the ratio of the two offices is exactly ½	conditional, 164.5
the ratio of the W and Z masses	open, 160.1

Confinement as the dark genre

Result	Grade	Box
Confinement as the dark-genre readability	conditional	171.1
The strong coupling as the gluon closed read	conditional	172.1
Nucleon structure as the star record census	conditional	174.1
The hadron spectrum as saturated joined webs	conditional	177.1
The quark offices as the star arm reads	**theorem**	181.5

At all 494 qutrit–qutrit links of the roster, no read survives at first order and the second-order read carries the readout. A hadron is a star: a junction qutrit of degree three carrying three quark bodies.



The proton as a seeded sink is an open result (175.1). Its quark content, two ups and a down, is a reading of its three docks, not a derivation.

Names from the literature are correspondences by name; each box is a statement about the networks, at the grade shown

Gravity as a statement about counting

Result	Grade	Box
Depth metric with the rung term	conditional	90.1
Curvature as the graph Laplacian washed over the ladder	conditional	90.2
The horizon at every rung as a seeded saturation	conditional	99.1
The abundances and the rate at the horizon	**theorem**	104.1
The cosmological term at the depth exponent	conditional	115.1

What Part VI builds

From counts alone, Part VI builds a metric from the depth of a web, curvature as a graph Laplacian, and a horizon at every rung. It proves the abundances and the rate at the horizon (Box 104.1), and it writes the cosmological term as an exact object at the depth exponent.

Mass is a read

Result	Grade	Box
The mass unit and the first power	conditional	185.1
The body-alone office rule	**theorem**	195.1
Generation as depth momentum	conditional	197.1
The unified mass readout	conditional	198.3
The Higgs as the closed hub read	conditional	200.1
The Higgs quartic as the ratio of offices	**theorem**	202.1

The Higgs quartic is an exact element of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ at every one of 10,123 Higgs-form sinks. No measured mass enters any derivation as an input.

Outside the Standard Model, mostly theorems

Thermodynamics

- The second law at the counting face (217.1)
- The phase transition as a change of sink class (221.1)
- The distillation theorem at the forward saturation (220.3)

Condensed matter

- The chain of copies and the periodic readout (224.1)
- The band at the periodic web (225.1)
- The winding of the wash holonomy, read as the Chern number (I.51.2–3)

Quantum information

- The closure of a web as a computation (230.1)
- The read as the maximal distillation (233.2)
- The holonomy of basis transport around a closed path (235.2)

All nine are theorems. Here the construction’s vocabulary and the literature’s come closest: both speak of qudits, entangled pairs and gates.

A SHARP TEST

$$1/\alpha_{\text{cand}} = 137.035\,999\,189\,566\,155\,029\,183\,744\ldots$$

The candidate fixes every digit in advance. Each further digit that measurements of α(0) reach is a sharper test, and continued agreement lends growing support to the axiom and the Photon Conjecture.

The exact form carries no uncertainty of its own. Measurements that settle today’s tension between rubidium and cesium recoil will test the next digits.

HOW TO CHECK IT

Everything is in one repository

The alpha computation

```
cd papers/alpha/lean
lake build
python3 audit.py
```

About two minutes and 2 GB. A Python script rebuilds the 172 records without Lean.

The compendium

Start at its overview: each Part’s opening and its graded statements. Its index lists every boxed statement with its grade, premises and Lean declarations.

Its Lean builds with 16 GB of memory, or with swap.

Find an error

Corrections and objections are welcome, by issue on the repository or by email.

The chain-theory repository on GitHub
alexwg@alexwg.org

SUMMARY

Five things to take away

One axiom	Finitely many maximally entangled pairs of qudits, some shared, known only by distilling them.
One cap	Certainty that leaves nothing open gives the read cap $d_1 d_2 \leq 6$, with two readable genres, one of them with an acting central charge.
One number	$(79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6})/31104$, computed with nothing to fix, and checked in Lean.
One conjecture	That it is $\alpha(0)$: ten significant figures of agreement, 0.6σ from CODATA 2022, with every further digit a sharper test.
A compendium	What else has been worked out, graded: 100 theorems, 109 conditional theorems and 198 open results, its correspondences with physics conjectured.

CONTACT

Dr. Alexander D. Wissner-Gross

alexwg@alexwg.org

https://www.alexwg.org

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BACKUP

The sector sizes from traces of the two exchanges

An **involution** S has $S^2 = 1$: its eigenvalues are ±1, and tr S = (number of +1 directions) − (number of −1 directions). X and Y commute, so the sector X = x, Y = y holds $\frac{1}{4}(172 + x\,\mathrm{tr}\,X + y\,\mathrm{tr}\,Y + xy\,\mathrm{tr}\,XY)$ records, the trace of $\frac{1}{4}(1 + xX)(1 + yY)$.

Swap rule: a swap's trace is the number of records it leaves in place, dim V on $V \otimes V$, and 0 on two edges it exchanges.

Block	X	Y	tr X	tr Y	tr XY
Kept edges, 73 records	−swap	1	−(8 + 3) = −11	73	−11
Swapped edges, 96 records	1	−swap	96	0	0
Commutators, 3 records	−1	−1	−3	−3	3
All 172			82	70	−8

$(+,+)$: $\frac{1}{4}(172 + 82 + 70 - 8) = 79$ · $(+,-)$: $\frac{1}{4}(172 + 82 - 70 + 8) = 48$ · $(-,+)$: $\frac{1}{4}(172 - 82 + 70 + 8) = 42$ · $(-,-)$: $\frac{1}{4}(172 - 82 - 70 - 8) = 3$

BACKUP

The Lean main theorem

```
theorem main :
    TwoPoint.valid objectWorld objectTwoPoint = true ∧
    fixesFar (objectWorld.dim objectTwoPoint.center) (objectWorld.dim objectTwoPoint.read1.2) = true ∧
    fixesFar (objectWorld.dim objectTwoPoint.read1.2) (objectWorld.dim objectTwoPoint.center) = false ∧
    (records (readEnd ...) (farEnd ...)).all (fun f => neZ (parity f)) = true ∧
    readout objectWorld objectTwoPoint = ([(1, 79), (2, 48), (3, 42), (6, 3)], 31104)
```

1. The object is a two-point configuration of the qutrit–qubit link.
2. Reads at the qutrit fix the qubit, and reads at the qubit do not fix the qutrit: the end rule.
3. Every one of the link’s records is an eigenvector of the exchange of the two reads.
4. The normalized readout is (79 + 48√2 + 42√3 + 3√6)/31104.

BACKUP

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